RADIAL UNIQUENESS OF THE NONZERO 1-POINTS OF J_0

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ABSTRACT. We answer Problem 2.45 in the Hayman–Lingham problem collection. If $z_0 \neq 0$, $J_0(z_0) = 1$, and $c > 1$, then $|J_0(cz_0)| > c^{7/25}$. Hence no two distinct nonzero 1-points of J_0 lie on the same open ray from the origin. The proof uses an exact, branch-consistent Laplace decomposition into growing and decaying components, and uniform estimates show that the separation persists along the outward ray and yields a positive radial logarithmic derivative. As an application, for distinct positive radii r_1, r_2 and $R > r_1 + r_2$, raywise uniqueness makes Volchkov's positive nonunit spectral-ratio exclusion automatic in the local planar spherical-mean theorem and gives the corresponding global L^1_{loc} result.

1. INTRODUCTION

Let J_0 denote the Bessel function of the first kind of order zero. Problem 2.45 in the Hayman–Lingham edition of Hayman's problem collection asks whether $J_0(z) = 1$ has at most one nonzero solution on each ray from the origin. Hayman and Lingham attribute the question to L. Zalcman and record that the available asymptotic information gives only finitely many solutions on any fixed ray [4, Problem 2.45].

We answer this question by proving the following stronger estimate.

Theorem A. If $z_0 \in \mathbb{C} \setminus \{0\}$ and $J_0(z_0) = 1$, then

$$|J_0(cz_0)| > c^{7/25} \qquad (c > 1).$$

Since $c^{7/25} > 1$ for $c > 1$, Theorem A answers Problem 2.45 immediately.

Corollary B. For every $\theta \in [0, 2\pi)$,

$$\#\{r > 0 : J_0(re^{i\theta}) = 1\} \leq 1.$$

Equivalently, no two distinct nonzero 1-points of J_0 lie on the same open ray from the origin.

2020 *Mathematics Subject Classification.* Primary 33C10; Secondary 30D20, 31A05, 35J05.

Key words and phrases. Bessel function, 1-point, radial uniqueness, radial escape, circular mean, two-radius theorem.

Helen W. J. Zhang was supported by the National Natural Science Foundation of China (Grant No. 12371327) and Hunan Provincial Natural Science Foundation of China (Grant No. 2026JJ50107).

The question also has a spectral interpretation in converse mean-value theorems. Delsarte and Lions showed that two fixed spherical mean identities force harmonicity when the corresponding averaging operators have no nonzero common fixed spectral mode [2]; the Euclidean formulation is given in Delsarte's lectures [1, Part III, §§3–4]. In dimension two the common spectral equations are

$$J_0(i\lambda r_1) = J_0(i\lambda r_2) = 1, \qquad \lambda \neq 0.$$

If $0 < r_1 < r_2$, $c = r_2/r_1$, and $z = i\lambda r_1$, these equations become $J_0(z) = J_0(cz) = 1$. Thus a positive nonunit spectral ratio occurs exactly when two nonzero 1-points lie on one open ray.

Volchkov's local planar classification separates the strict geometric condition $R > r_1 + r_2$ from the exclusion of a Bessel root-ratio set [5]. Theorem A removes the positive nonunit part of this spectral obstruction, yielding the following consequence.

Corollary C. Let $B_R := \{x \in \mathbb{R}^2 : |x| < R\}$, let r_1, r_2 be distinct positive radii, and let $R > r_1 + r_2$. If $f \in L^1_{\mathrm{loc}}(B_R)$ satisfies the normalized circular mean identities at r_1 and r_2 almost everywhere on their natural domains, then $\Delta f = 0$ in $\mathcal{D}'(B_R)$. In the full-space case, if $f \in L^1_{\mathrm{loc}}(\mathbb{R}^2)$ satisfies the two identities almost everywhere on $\mathbb{R}^2$, then $\Delta f = 0$ in $\mathcal{D}'(\mathbb{R}^2)$.

A radial energy identity and an elementary zero-free disk first localize every nonzero 1-point. On the principal branch of K_0, a Laplace representation then gives an exact, branch-consistent decomposition of J_0 into growing and decaying components. Uniform amplitude estimates separate the two components at each principal-sector 1-point, and this separation persists along the whole outward ray. The resulting lower bound for the radial logarithmic derivative implies Theorem A; evenness and conjugation symmetry give the global statement.

This paper was submitted on 30 August 2026. Ferudun's preprint [9], dated 3 October 2026, presents a computer-assisted proof of the same ray-wise uniqueness statement. Both arguments separate growing and decaying Bessel components. Ferudun combines explicit Hankel bounds, contraction and Rouché arguments, certified root counting, and ordering of root arguments. Our proof uses an exact K_0 Laplace decomposition and uniform amplitude and derivative estimates propagated along outward rays to obtain the quantitative estimate of Theorem A, without enumerating the 1-points.

Section 2 gives the localization. Sections 3 and 4 establish the exact decomposition and the uniform component bounds. Section 5 proves radial escape and answers Problem 2.45. Section 6 proves the two-radius consequences and compares them with Volchkov's local theorem.

The results proved here have been formalized in Lean 4 using mathlib [6, 7]. The source code, statement correspondence, and reproducible build instructions are available in the accompanying GitHub repository.

2. Preliminary localization of the nonzero 1-points

We use the defining series

$$J_0(z) := \sum_{m=0}^{\infty} \frac{(-1)^m z^{2m}}{4^m (m!)^2}, \qquad z \in \mathbb{C}.$$

Set

$$F(w) := \sum_{m=0}^{\infty} \frac{w^m}{(m!)^2}, \qquad Q(w) := \sum_{n=0}^{\infty} \frac{w^n}{((n+1)!)^2}.$$

Then

$$J_0(z) = F\left(-\frac{z^2}{4}\right), \qquad F(w) - 1 = wQ(w). \tag{2.1}$$

Lemma 2.1. *If $w \neq 0$ and $\Re w \geq 0$, then $|F(w)| > 1$.*

Proof. Termwise differentiation gives $wF''(w) + F'(w) = F(w)$. Put $h(s) = F(sw)$ and $Y(s) = |h(s)|^2$. Then $sh'' + h' = wh$, and integration of $(sY')' = 2(s|h'|^2 + \Re w\,|h|^2)$ gives

$$sY'(s) = 2\int_0^s \left(u|h'(u)|^2 + \Re w\,|h(u)|^2\right) du. \tag{2.2}$$

The integrand is nonnegative for $u \geq 0$. Since $h'(0) = w \neq 0$, its first term is positive for all sufficiently small $u > 0$. Thus $Y'(s) > 0$ for every $s > 0$, and $|F(w)|^2 = Y(1) > Y(0) = 1$. □

Lemma 2.2. *For every $x \in \mathbb{R} \setminus \{0\}$, one has $|J_0(x)| < 1$.*

Proof. Put $q = x^2/4 > 0$ and $h(s) = F(-qs)$. The equation for F gives $sh'' + h' = -qh$. Hence the energy

$$E(s) := |h(s)|^2 + \frac{s}{q}|h'(s)|^2 \qquad \text{satisfies} \qquad E'(s) = -\frac{1}{q}|h'(s)|^2.$$

Since $h'(0) = -q \neq 0$, integration on $[0,1]$ gives $E(1) < E(0) = 1$. Therefore $|J_0(x)|^2 = |h(1)|^2 \leq E(1) < 1$. □

Lemma 2.3. *The function Q has no zero in the disk $|w| < 4$.*

Proof. Write $w = 4u$, where $|u| < 1$, and put $b_n = 4^n/((n+1)!)^2$. The sequence (b_n) is nonnegative and summable, with $b_0 = 1$ and $b_{n+1}/b_n = 4/(n+2)^2 \leq 1$. Set $d_n = b_n - b_{n+1} \geq 0$. Absolute convergence and telescoping give

$$(1-u)Q(4u) = 1 - \sum_{n=0}^{\infty} d_n u^{n+1}, \qquad \sum_{n=0}^{\infty} d_n = 1. \tag{2.3}$$

Moreover,

$$\left|\sum_{n=0}^{\infty} d_n u^{n+1}\right| \leq |u| \sum_{n=0}^{\infty} d_n = |u| < 1.$$

Thus $(1-u)Q(4u) \neq 0$, and hence $Q(w) \neq 0$. □

Proposition 2.4. *If $J_0(z) = 1$ and $z \neq 0$, then $\Re(z^2) > 0$, $z \notin \mathbb{R}$, $|z| \geq 4$.*

Proof. Let $w = -z^2/4$. If $\Re(z^2) \leq 0$, then $w \neq 0$ and $\Re w \geq 0$, so Lemma 2.1 gives $|J_0(z)| = |F(w)| > 1$, a contradiction. Lemma 2.2 excludes nonzero real z. Finally, (2.1) gives $wQ(w) = 0$. Since $w \neq 0$, Lemma 2.3 yields $|w| \geq 4$, and hence $|z| \geq 4$. □

Thus every nonzero 1-point lies in one of the four open sectors adjacent to the real axis. By evenness and conjugation symmetry, it will later be enough to work in $0 < \arg z < \pi/4$.

3. An exact Laplace decomposition

Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$. On Ω we use the principal logarithm $\operatorname{Log} w = \log|w| + i \operatorname{Arg} w$, where $-\pi < \operatorname{Arg} w < \pi$, and define all half-integral powers through this branch. Let K_0 denote the modified Bessel function of the second kind of order zero, with its principal branch. Put

$$d\mu(t) := \frac{e^{-t}t^{-1/2}}{\sqrt{\pi}}\,dt, \qquad \int_0^\infty d\mu(t) = 1, \qquad \int_0^\infty t\,d\mu(t) = \frac{1}{2}.$$

For $w \in \Omega$, define

$$\mathcal{I}(w) := \int_0^\infty \left(1 + \frac{t}{2w}\right)^{-1/2} d\mu(t), \qquad \mathcal{K}(w) := \sqrt{\frac{\pi}{2w}}e^{-w}\mathcal{I}(w).$$

Proposition 3.1. *The functions $\mathcal{I}$ and $\mathcal{K}$ are holomorphic on Ω,*

$$\mathcal{I}'(w) = \frac{1}{4w^2}\int_0^\infty t\left(1 + \frac{t}{2w}\right)^{-3/2} d\mu(t),$$

and $\mathcal{K}(w) = K_0(w)$ for every $w \in \Omega$.

Proof. For $w \in \Omega$ and $t > 0$, the point $1 + t/(2w)$ does not lie on $(-\infty, 0]$: otherwise $1 + t/(2w) = -s$ for some $s \geq 0$, and then $w = -t/[2(1+s)] \in (-\infty, 0)$. Thus the integrand is holomorphic in w on Ω.

Let $L \Subset \Omega$ be nonempty and compact. Compactness gives positive constants m_L, M_L, η_L such that, for $w \in L$ and $0 \leq s \leq 1$,

$$m_L \leq |w| \leq M_L, \qquad |2(1-s)w + s| \geq \eta_L.$$

Indeed, a zero of the latter expression would force either $s = 1$ and $1 = 0$, or $w = -s/[2(1-s)] \in (-\infty, 0]$. Taking $s = t/(1+t)$ shows that

$$\left|1 + \frac{t}{2w}\right| \geq c_L := \frac{\eta_L}{2M_L} > 0 \qquad (w \in L,\ t \geq 0).$$

The kernel is therefore bounded on L by $c_L^{-1/2}$. Its derivative is

$$\frac{\partial}{\partial w}\left(1 + \frac{t}{2w}\right)^{-1/2} = \frac{t}{4w^2}\left(1 + \frac{t}{2w}\right)^{-3/2},$$

whose modulus is at most $tc_L^{-3/2}/(4m_L^2)$. The mass and first moment of μ now justify differentiation under the integral sign. This proves the holomorphy of

$\mathcal{I}$ and its derivative formula; the defining prefactor then gives the holomorphy of $\mathcal{K}$.

For $w > 0$, the standard representation [3, 10.32.9]

$$K_0(w) = \int_0^\infty e^{-w\cosh u}\,du$$

and the substitution $t = w(\cosh u - 1)$ give

$$K_0(w) = e^{-w}\int_0^\infty \frac{e^{-t}}{\sqrt{t(2w+t)}}\,dt = \mathcal{K}(w).$$

The principal branch of K_0 is holomorphic on Ω [3, Sec. 10.25(ii)]. Since Ω is connected, the identity theorem yields $\mathcal{K} = K_0$ throughout Ω. □

Let $\mathbb{H}_+ = \{z \in \mathbb{C} : \Re z > 0\}$. For $z \in \mathbb{H}_+$, define

$$A_\pm(z) := \int_0^\infty \left(1 \pm \frac{it}{2z}\right)^{-1/2} d\mu(t).$$

Proposition 3.2. *The functions $A_\pm$ are holomorphic on $\mathbb{H}_+$ and satisfy*

$$A'_\pm(z) = \pm\frac{i}{4z^2}\int_0^\infty t\left(1 \pm \frac{it}{2z}\right)^{-3/2} d\mu(t). \tag{3.1}$$

Moreover, for every $z \in \mathbb{H}_+$,

$$J_0(z) = \frac{e^{i(z-\pi/4)}A_+(z) + e^{-i(z-\pi/4)}A_-(z)}{\sqrt{2\pi z}}, \tag{3.2}$$

where the square root is taken on the principal branch.

Proof. Since $-iz, iz \in \Omega$ for $z \in \mathbb{H}_+$, one has $A_+(z) = \mathcal{I}(-iz)$ and $A_-(z) = \mathcal{I}(iz)$. Holomorphy and (3.1) follow from Proposition 3.1 and the chain rule.

Write $z = re^{i\theta}$ with $-\pi/2 < \theta < \pi/2$. Then

$$\operatorname{Log}(iz) = \operatorname{Log} z + \frac{\pi i}{2}, \qquad \operatorname{Log}(-iz) = \operatorname{Log} z - \frac{\pi i}{2},$$

and therefore

$$(iz)^{-1/2} = e^{-\pi i/4}z^{-1/2}, \qquad (-iz)^{-1/2} = e^{\pi i/4}z^{-1/2}.$$

Proposition 3.1 gives

$$K_0(-iz) = \sqrt{\frac{\pi}{2}}\,e^{\pi i/4}z^{-1/2}e^{iz}A_+(z),$$

$$K_0(iz) = \sqrt{\frac{\pi}{2}}\,e^{-\pi i/4}z^{-1/2}e^{-iz}A_-(z).$$

The standard connection formula [3, 10.27.9], specialized to order zero, is

$$\pi i J_0(z) = K_0(-iz) - K_0(iz), \qquad \Re z > 0,$$

with both values taken on the principal branch. Substitution and the identities

$$\frac{1}{\pi i}\sqrt{\frac{\pi}{2}}e^{\pi i/4} = \frac{e^{-\pi i/4}}{\sqrt{2\pi}}, \qquad -\frac{1}{\pi i}\sqrt{\frac{\pi}{2}}e^{-\pi i/4} = \frac{e^{\pi i/4}}{\sqrt{2\pi}}$$

yield (3.2). □

4. Control of the two components

Let $z = x + iy = \rho e^{i\theta}$, where $\rho > 0$ and $0 \le \theta \le \pi/4$, and put $\kappa = 2^{-9/4}$.

Proposition 4.1. *In this closed sector,*

$$|A_\pm(z) - 1| \le \frac{\kappa}{\rho}, \qquad |A'_\pm(z)| \le \frac{\kappa}{\rho^2}. \tag{4.1}$$

If $\rho \ge 4$, then

$$\frac{15}{16} < |A_\pm(z)| < \frac{17}{16}, \qquad \left| z\frac{A'_\pm(z)}{A_\pm(z)} \right| < \frac{1}{15}. \tag{4.2}$$

Proof. Since $x \ge y \ge 0$ and $x > 0$, completing squares in Cartesian coordinates gives, for $t \ge 0$ and $0 \le \sigma \le 1$,

$$\left|1 + \sigma\frac{it}{2z}\right|^2 \ge 1, \qquad \left|1 - \sigma\frac{it}{2z}\right|^2 \ge \frac{1}{2}.$$

For $t > 0$ and $\sigma > 0$, these points have respectively positive and negative imaginary parts; the zero endpoints equal 1. Thus both line segments avoid the principal cut. Along either segment, the derivative of $(1+\sigma\xi)^{-1/2}$, where $\xi = \pm it/(2z)$, has modulus at most $2^{3/4}|\xi|/2$. The mean-value estimate therefore gives

$$\left| \left(1 \pm \frac{it}{2z}\right)^{-1/2} - 1 \right| \le \frac{2^{3/4}t}{4\rho}.$$

Using $\int t\, d\mu(t) = 1/2$ proves the first estimate in (4.1). The derivative formula (3.1) and the same geometric bounds give the second.

If $\rho \ge 4$, then $\kappa/\rho < 1/16$. The triangle and reverse triangle inequalities yield $15/16 < |A_\pm(z)| < 17/16$; in particular, $A_\pm(z) \ne 0$. Finally,

$$\left| z\frac{A'_\pm(z)}{A_\pm(z)} \right| < \frac{1}{15}.$$

□

Let

$$\mathcal{U} := \{z \in \mathbb{H}_+ : A_+(z)A_-(z) \ne 0\}.$$

Since A_+A_- is holomorphic on $\mathbb{H}_+$, the set $\mathcal{U}$ is open. On $\mathcal{U}$, define

$$\delta(z) := e^{2i(z-\pi/4)}\frac{A_+(z)}{A_-(z)}. \tag{4.3}$$

Then, throughout $\mathcal{U}$,

$$J_0(z) = \frac{e^{-i(z-\pi/4)}A_-(z)}{\sqrt{2\pi z}}\big(1+\delta(z)\big). \tag{4.4}$$

Proposition 4.2. *Suppose $J_0(z) = 1$ and $z = x+iy = \rho e^{i\theta}$ with $0 < \theta < \pi/4$. Then $z \in \mathcal{U}$, and*

$$y > \frac{7}{5}, \qquad e^{-2y} < \frac{1}{16}, \qquad |\delta(z)| < \frac{17}{240}, \qquad \rho e^{-2y} < \frac{11}{50}.$$

Proof. By Proposition 2.4, $\rho \geq 4$, so Proposition 4.1 gives $z \in \mathcal{U}$. Using $J_0(z) = 1$, (3.2), and (4.2), we obtain

$$\sqrt{8\pi} \leq \sqrt{2\pi\rho} \leq \frac{17}{16}(e^y + e^{-y}) = \frac{17}{8}\cosh y.$$

Hence $\cosh y \geq (8/17)\sqrt{8\pi} > 11/5$. Together with $\cosh(7/5) < 11/5$ and $e^{14/5} > 16$, this gives $y > 7/5$ and $e^{-2y} < 1/16$, since $y > 0$.

By (4.3) and (4.2),

$$|\delta(z)| = e^{-2y}\left|\frac{A_+(z)}{A_-(z)}\right| < \frac{1}{16}\frac{17}{15} = \frac{17}{240}.$$

Multiplying the preceding bound for $\sqrt{2\pi\rho}$ by e^{-y} gives

$$\sqrt{2\pi\rho}\, e^{-y} \leq \frac{17}{16}(1+e^{-2y}) \leq \left(\frac{17}{16}\right)^2.$$

Squaring and using $\pi > 3$, we conclude that

$$\rho e^{-2y} \leq \frac{(17/16)^4}{2\pi} < \frac{(17/16)^4}{6} < \frac{11}{50}. \qquad \square$$

Proposition 4.2 is the point at which the increasing component becomes uniformly dominant. The next section shows that this dominance persists along the whole outward ray.

5. Radial escape

Proposition 5.1. *Suppose $J_0(z_1) = 1$ and $z_1 = x_1 + iy_1 = \rho_1 e^{i\theta}$ with $\rho_1 > 0$ and $0 < \theta < \pi/4$. For $\tau \geq 1$, put $z_\tau = \tau z_1$. Then*

$$\tau\frac{d}{d\tau}\log|J_0(z_\tau)| > \frac{7}{25} \qquad (\tau \geq 1), \tag{5.1}$$

and hence

$$|J_0(cz_1)| > c^{7/25} \qquad (c > 1). \tag{5.2}$$

Proof. Let $\rho_\tau = |z_\tau| = \tau\rho_1$ and $y_\tau = \Im z_\tau = \tau y_1$. The argument is fixed and $\rho_\tau \geq \rho_1 \geq 4$, so Proposition 4.1 applies along the entire ray.

By Proposition 4.2, $\rho_1 e^{-2y_1} < 11/50$. Since

$$\rho_\tau e^{-2y_\tau} = \rho_1 e^{-2y_1}\,\tau e^{-2(\tau-1)y_1}$$

and $\log\tau \le \tau - 1$, we have $\tau e^{-2(\tau-1)y_1} \le 1$. Hence

$$\rho_\tau e^{-2y_\tau} < \frac{11}{50}.$$

Also $e^{-2y_\tau} < 1/16$. Therefore

$$|\delta(z_\tau)| < \frac{17}{240}, \qquad \rho_\tau|\delta(z_\tau)| < \frac{187}{750}. \tag{5.3}$$

Thus $1 + \delta(z_\tau) \neq 0$, while $A_-(z_\tau) \neq 0$ by (4.2). The factorization (4.4) implies

$$J_0(z_\tau) \neq 0 \qquad (\tau \ge 1). \tag{5.4}$$

Since J_0 is entire, (5.4) and the identity $\log|g| = \frac{1}{2}\log|g|^2$ for $g \neq 0$ show that $\log|J_0(\tau z_1)|$ is differentiable near each $\tau \ge 1$, including $\tau = 1$. Logarithmic differentiation of (4.4) along the ray gives

$$\tau\frac{d}{d\tau}\log|J_0(z_\tau)| = y_\tau - \frac{1}{2} + \Re\left(z_\tau\frac{A_-'(z_\tau)}{A_-(z_\tau)} + z_\tau\frac{\delta'(z_\tau)}{1+\delta(z_\tau)}\right). \tag{5.5}$$

Differentiation of (4.3) gives

$$z\delta'(z) = \delta(z)\left(2iz + z\frac{A_+'(z)}{A_+(z)} - z\frac{A_-'(z)}{A_-(z)}\right). \tag{5.6}$$

Using (5.6), (4.2), (5.3), and the triangle inequality, we obtain

$$\left|z_\tau\frac{A_-'(z_\tau)}{A_-(z_\tau)} + z_\tau\frac{\delta'(z_\tau)}{1+\delta(z_\tau)}\right| < \frac{1}{15} + \frac{2(187/750) + (2/15)(17/240)}{1 - 17/240} < \frac{31}{50}.$$

Substitution into (5.5), together with $y_\tau \ge y_1 > 7/5$, proves (5.1). Finally, put

$$H(\tau) := \log|J_0(\tau z_1)| - \frac{7}{25}\log\tau.$$

The bound (5.1) gives $H'(\tau) > 0$ for $\tau \ge 1$. Since $H(1) = 0$, the mean value theorem yields $H(c) > 0$ for $c > 1$, which proves (5.2). □

Theorem 5.2. *If $z_0 \in \mathbb{C} \setminus \{0\}$ and $J_0(z_0) = 1$, then*

$$|J_0(cz_0)| > c^{7/25} \qquad (c > 1). \tag{5.7}$$

Proof. The defining series gives $J_0(-z) = J_0(z)$ and $J_0(\overline{z}) = \overline{J_0(z)}$. Write $z_0 = x + iy$. By Proposition 2.4, $0 < |y| < |x|$. Thus $w = |x| + i|y|$ lies in $0 < \arg w < \pi/4$ and is one of $\pm z_0, \pm\overline{z_0}$. Hence $J_0(w) = 1$ and, for every real $c > 0$, $|J_0(cw)| = |J_0(cz_0)|$. Proposition 5.1 applied to w proves (5.7). □

Corollary 5.3. *For every $\theta \in [0, 2\pi)$, $\#\{r > 0 : J_0(re^{i\theta}) = 1\} \le 1$.*

Proof. If $0 < r_1 < r_2$ gave two solutions on the same ray, then with $c = r_2/r_1 > 1$ and $z_1 = r_1 e^{i\theta}$, Theorem 5.2 would give $|J_0(r_2 e^{i\theta})| = |J_0(cz_1)| > 1$, a contradiction. □

6. Two-radius consequences

For $R > 0$, let $B_R = \{x \in \mathbb{R}^2 : |x| < R\}$. If $f \in L^1_{\mathrm{loc}}(B_R)$ and $0 < r < R$, define the normalized circular mean

$$M_r f(x) := \frac{1}{2\pi} \int_0^{2\pi} f\big(x + r(\cos t, \sin t)\big)\, dt$$

for almost every $x \in B_{R-r}$. Fubini's theorem shows that this defines an $L^1_{\mathrm{loc}}(B_{R-r})$ function independently of the chosen representative of f.

The full-space conditional two-radius theorem goes back to Delsarte and Lions [2, 1]. Volchkov later obtained a sharp local formulation that separates the spectral obstruction from the geometric condition on R, r_1, r_2 [5, Theorem 1]. In dimension two, define

$$Z := \{z \in \mathbb{C} \setminus \{0\} : J_0(z) = 1\}, \qquad \mathcal{A} := \left\{ \frac{\alpha}{\beta} : \alpha, \beta \in Z \right\}.$$

Proposition 6.1. *For positive radii r_1, r_2, the following are equivalent:*

1. *$r_1/r_2 \in \mathcal{A}$;*
2. *there exists $\lambda \in \mathbb{C} \setminus \{0\}$ such that*

$$J_0(i\lambda r_1) = J_0(i\lambda r_2) = 1.$$

Proof. If $\alpha/\beta = r_1/r_2$ with $\alpha, \beta \in Z$, set $\lambda = -i\alpha/r_1 = -i\beta/r_2$. Conversely, a common nonzero spectral point gives $\alpha = i\lambda r_1$, $\beta = i\lambda r_2$, and $\alpha/\beta = r_1/r_2$. □

A common nonzero spectral parameter gives the following counterexample.

Proposition 6.2. *If the equivalent conditions in Proposition 6.1 hold, then there is a real-valued C^∞ function on $\mathbb{R}^2$ that is not harmonic and satisfies both circular mean identities.*

Proof. For $F_\lambda(x_1, x_2) = e^{\lambda x_1}$,

$$M_r F_\lambda = F_\lambda \frac{1}{2\pi} \int_0^{2\pi} e^{\lambda r \cos t}\, dt = J_0(i\lambda r) F_\lambda.$$

Thus the two common spectral identities imply $M_{r_1} F_\lambda = M_{r_2} F_\lambda = F_\lambda$, whereas $\Delta F_\lambda = \lambda^2 F_\lambda$. At the origin this Laplacian equals $\lambda^2 \neq 0$, so at least one of $\Re F_\lambda$ and $\Im F_\lambda$ is a real-valued smooth nonharmonic example satisfying the same identities. □

Theorem 5.2 gives

$$(\mathcal{A} \cap (0, \infty)) \setminus \{1\} = \varnothing. \tag{6.1}$$

Indeed, if $\alpha/\beta = c > 0$ with $c \neq 1$, then either $c > 1$ or $c^{-1} > 1$, and two distinct points of Z would lie on one open ray.

We use the following dimension-two specialization of Volchkov's local theorem [5, Theorem 1(i)].

Theorem 6.3. *Let $r_1, r_2 > 0$, let $R > r_1 + r_2$, and let $f \in L^1_{\mathrm{loc}}(B_R)$. Suppose*

$$M_{r_j} f = f \quad \text{a.e. on } B_{R-r_j}, \qquad j = 1, 2.$$

If $r_1/r_2 \notin \mathcal{A}$, then $\Delta f = 0$ in $\mathcal{D}'(B_R)$.

Corollary 6.4. *Let $r_1 \neq r_2$ be positive, let $R > r_1 + r_2$, and let $f \in L^1_{\mathrm{loc}}(B_R)$. If*

$$M_{r_j} f = f \quad \text{a.e. on } B_{R-r_j}, \qquad j = 1, 2,$$

then $\Delta f = 0$ in $\mathcal{D}'(B_R)$.

Proof. The ratio r_1/r_2 is positive and different from 1. By (6.1), it does not belong to $\mathcal{A}$. The conclusion follows from Theorem 6.3. □

Corollary 6.5. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^2)$, and let $r_1 \neq r_2$ be positive. If*

$$M_{r_1} f = M_{r_2} f = f \quad \text{a.e. on } \mathbb{R}^2,$$

then $\Delta f = 0$ in $\mathcal{D}'(\mathbb{R}^2)$. Consequently, f agrees almost everywhere with a smooth harmonic function. If f is continuous, then f itself is harmonic and has the circular mean-value property at every positive radius.

Proof. For every $R > r_1 + r_2$, Corollary 6.4 applies to the restriction of f to B_R. Hence $\Delta f = 0$ in every such ball and therefore in $\mathcal{D}'(\mathbb{R}^2)$. Weyl's lemma gives a smooth harmonic representative. The final assertion follows from continuity and the usual mean-value theorem for harmonic functions. □

Declarations

Use of generative AI. The Lean formalization was completed using the local Prove2Me workflow [8] with GPT-6.1 Sol. Generative AI tools also assisted with mathematical exploration, language, and LaTeX editing. The authors reviewed the resulting material and take responsibility for the mathematical content and its correspondence with the formal statements.

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